



LaSalle County Station
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10 CFR 50.73

RA21-009

February 18, 2021

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, Unit 2
Renewed Facility Operating License No. NPF-18
NRC Docket No. 50-374

Subject: Licensee Event Report 2021-001-00, High Pressure Core Spray
Inoperable due to Water Leg Pump Breaker Trip

In accordance with 10 CFR 50.73(a)(2)(v)(D), Exelon Generation Company, LLC (EGC) is submitting Licensee Event Report (LER) Number 2021-001-00 for LaSalle County Station, Unit 2.

There are no regulatory commitments in this letter. Should you have any questions concerning this report, please contact Mr. Daniel Mearhoff, Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

A handwritten signature in black ink that reads "Phil Hansett".

Phil Hansett
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk alt: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

LaSalle County Station, Unit 2

2. Docket Number

05000 374

3. Page

1 OF 3

4. Title

High Pressure Core Spray Inoperable due to Water Leg Pump Breaker Cubicle Motor Contactor

5. Event Date

Month	Day	Year
12	23	2020

6. LER Number

Year	Sequential Number	Revision No.
2021	- 001 -	00

7. Report Date

Month	Day	Year
02	18	2021

8. Other Facilities Involved

Facility Name	Docket Number
NA	05000
Facility Name	Docket Number
NA	05000

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Amy McMartin, Site Engineering Director

Phone Number (Include area code)

(815) 415-3800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	BG	CNTR	G080	Y	NA	NA	NA	NA	NA

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year
NA	NA	NA

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On December 23, 2020, the Unit 2 high pressure core spray (HPCS) water leg pump (WLP) tripped off due to a fault within its breaker cubicle. The HPCS WLP is a support component required to maintain the HPCS system operable. With the loss of the WLP, the HPCS system was declared inoperable. The unit entered applicable technical specification required actions to immediately verify that the reactor core isolation cooling (RCIC) system was operable and to restore the HPCS system to operable within 14 days.

This condition was previously reported under emergency notification system (ENS) 55046 as a single-train system inoperability that could have prevented the fulfillment of a safety function used to mitigate an accident. The condition is being reported under 10CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented fulfillment of a safety function. The system was restored to operable following replacement of the affected HPCS WLP breaker cubicle control power transformer and motor contactor. The cause was age-related failure of the breaker cubicle motor contactor.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 2	05000- 374	2021	- 001	- 00

NARRATIVE**Plant and System Identification**

LaSalle County Station Unit 2 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power. The affected system was the Division 3 high-pressure core spray (HPCS). In accident situations, the HPCS system provides cooling to the reactor core to prevent excessive fuel cladding temperatures following a break in the nuclear system piping.

Condition Prior to Event

Unit(s): 2 Date: December 23, 2020 Time: 06:53 CST
Reactor Mode(s): 1 Mode(s) Name: Power Operation Power Level: 100 percent

Description

On December 23, 2020, the main control room operating crew received an alarm for the Unit 2 HPCS water leg pump (WLP) having tripped off, followed shortly by a ground alarm for the associated electrical circuitry that cleared with 15 seconds. Field operators subsequently reported an acrid order in the Division 3 switchgear room and subsequently found the breaker for the HPCS WLP in the closed position but the breaker's cubicle was warm to the touch. The operating shift crew removed the HPCS WLP from service, declared the HPCS system inoperable, and entered applicable technical specifications (TS) 3.5.1 required action (RA) B.1 to immediately verify by administrative means that the reactor core isolation cooling (RCIC) system was operable, and TS 3.5.1 RA B.2 to restore the HPCS system to operable within 14 days. The operating shift established robust barriers to administratively protect the RCIC system following discovery through the subsequent restoration of the HPCS system inoperability period. The Unit 2 HPCS system was restored to operable the same day at 20:18 CST, after replacement of the WLP breaker cubicle control power transformer and motor contactor and post-maintenance testing.

Cause

Initial investigation identified that the WLP motor starter (2AP79E-2C) contactor and control power transformer were damaged. The cause was age-related failure of the contactor. The failure analysis found the motor starter damage had resulted from a heating event that began inside the contactor coil which eventually led to an open circuit in the coil. The increased current flow impacted the control power transformer, which overheated and failed.

Reportability and Safety Analysis

The HPCS system is part of the Division 3 emergency core cooling system (ECCS). In accident situations, HPCS provides cooling to the reactor core to prevent excessive fuel cladding temperatures following a break in the nuclear system piping. A secondary purpose of HPCS is to supply cooling water to the reactor pressure vessel as a backup to the RCIC system if the main condenser and feedwater were unavailable. The HPCS WLP is a support system required to maintain the HPCS system operable. With the loss of the HPCS WLP, HPCS was declared inoperable and the operating shift entered applicable TS RA statements.

This condition could have prevented the HPCS system, a single train safety system, from performing its design function. Therefore, this condition is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of the safety function of structures or systems that are required to mitigate the

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LaSalle County Station, Unit 2	05000- 374	2021	- 001	- 00

NARRATIVE

consequences of an accident. The event was previously communicated to the NRC under emergency notification system (ENS) 55046 in accordance with 10CFR 50.72(b)(3)(v)(D).

This condition was a safety system functional failure (SSFF) as defined in accordance with NEI 99-02, "Regulatory Assessment Performance Indicator Guideline." The support system WLP could have prevented the HPCS system from performing its design function.

Corrective Actions

Immediate corrective actions taken in response to the condition were:

- Declared the Unit 2 HPCS system inoperable on December 23, 2020 at 06:53 CST
- Completed maintenance and post-maintenance testing at 19:48 CST
- Declared Unit 2 HPCS operable and exited associated TS 3.5.1 required actions at 20:18 CST
- Performed failure analysis for the failed components to identify any additional actions

The extent of condition for potential similar failure mechanisms was determined to exist among the seven continuously energized contactors in each of the Unit 1 and Unit 2 Division 3 (HPCS) motor control centers (MMCs), a total of 14 contactors. A corrective action assignment was made to evaluate further for contactor replacements.

Previous Occurrences

There have been no similar events involving contactor faults for safety related systems within the previous three-year period. However, during that period there has been a previous event involving a breaker in the Unit 1 HPCS system MCC bus bar and another event involving a unit crosstie breaker.

- 373-2018-002-00, *Damaged Bus Bar Identified Potentially Affecting High Pressure Core Spray System*
Station personnel identified damage of the bus bar at the breaker that supplies the Unit 1 HPCS diesel generator auxiliaries. The degradation of the bus bar may have prevented the Division 3 DC battery charger from performing its function, which could have prevented the HPCS system from performing its design safety function. The condition of the MCC bus bar was the result of a degraded connection between the "A" phase bus bar and the "A" phase bus clip (stab) associated with the breaker cubicle 2B. The apparent cause of the degraded bus bar to clip connection was due to relaxation of the bucket clip over time.
- 374-2019-001-00, *Safety Related Electrical Bus Undervoltage Results in Valid Actuation of the Common Emergency Diesel Generator*
While shut down for a scheduled refueling outage, Unit 2 Division 1 safety-related electrical bus (241Y) tripped during surveillance testing. The trip of the electrical bus resulted in a valid undervoltage actuation signal to the common emergency diesel generator, causing it to start and load to bus. The cause was attributed to the crosstie breaker having a degraded lug on the terminal between its Unit 2 A-phase current transformer and the Unit 1 breaker differential current relay.

Component Failure Data

Device: 2AP79E-2C, HPCS WLP MCC
Manufacturer: General Electric [G080]
Component Type: Contactor [CNTR]
Model: CR206C0